Machine Learning 1 – Exercise 4

Machine Learning for Computer Vision
TU Dresden

due November 9 – 14, 2026

Deciding with linear functions: Logistic regression

a) In the lecture notes[1], derive (20) from (18) using (19), (7) and (8)

b) Prove that the objective function φ of the l_2-regularized logistic regression problem (20) in the lecture notes[1] is convex in θ.

$$\varphi(\theta) = \sum_{s \in S} \left(-y_s \langle \theta, x_s \rangle + \ln \left(1 + e^{\langle \theta, x_s \rangle} \right) \right) + \frac{1}{2\sigma^2} \|\theta\|_2^2 \tag{1}$$

c) Compute the gradient $\nabla_\theta \varphi$.

d) The provided code implements a flower segmentation task using logistic regression. Here, we assume $\sigma \to \infty$, i.e., we ignore the regularizer. The approach here is to look at a (k, k)-neighborhood of a pixel p in order to predict whether p belongs to the foreground or background.

Fill in the gaps in the provided code and implement gradient descent with a fixed learning rate $\lambda \in \mathbb{R}$, i.e., we update parameters according to

$$\theta \leftarrow \theta - \lambda \nabla_\theta \varphi \ . \tag{2}$$

[1] `https://mlcv.inf.tu-dresden.de/courses/26-winter/ml1/03-supervised-learning-linear.pdf`